



THE HILLS GRAMMAR SCHOOL
TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION
2003
MATHEMATICS
EXTENSION 1

Time Allowed: Two hours (plus 5 minutes reading time)

Teacher Responsible: Mr D Price

SPECIAL INSTRUCTIONS:

- This paper contains 7 questions. ALL questions to be attempted.
- ALL questions are of equal value.
- ALL necessary working should be shown in every question in the booklets provided.
- Start each question in a new booklet.
- A table of standard integrals is supplied at the back of this paper.
- Board approved calculators may be used.
- Hand up your paper in ONE bundle, together with this question paper.
- ALL HSC course outcomes are being assessed in this task. The Course Outcomes are listed on the back of this sheet.

Students are advised that this is a Trial Examination only and cannot in any way guarantee the content or the format of the Higher School Certificate Examination.

Question One

Marks

- (a) Evaluate $\int_0^{2\sqrt{3}} \frac{dx}{4+x^2}$ 2
- (b) Differentiate $\cos^3 x$ 2
- (c) Find the point which divides the line joining (4, 6) to (13, 5) externally in the ratio 4:1 2
- (d) Write down the equation of the vertical asymptote of $y = \frac{2x}{3x-1}$ 1
- (e) Solve for x : $\frac{3}{x+5} \leq 1$ 2
- (f) Evaluate $\int_0^{\frac{1}{\sqrt{2}}} \frac{2x^3}{\sqrt{1-x^4}} dx$ using the substitution $u = x^4$ 3

Question Two (Start a NEW booklet)

Marks

(a) Evaluate $\lim_{x \rightarrow 0} \frac{\tan 2x}{4x}$

1

(b) Solve the equation

$$\sin \theta + \sqrt{3} \cos \theta = 1 \text{ for } 0 \leq \theta \leq 2\pi$$

4

(c) Air is being pumped into a spherical balloon at the rate of $450\text{cm}^3 \text{ s}^{-1}$. Calculate the rate at which the radius of the balloon is increasing at the instant when the radius reaches 15cm. $\left[V = \frac{4}{3} \pi r^3 \right]$

3

(d) Let $f(x) = \cos x - \ln x$

4

(i) Show that a root to $f(x) = 0$ lies between 0.5 and 1.5.

(ii) Starting with a value of $x = 1$, use one application of Newton's method to find a better approximation to this root of $f(x) = 0$.

Question Three (Start a NEW booklet)

Marks

(a) The region R is bounded by the curve $y = \cos x$, $x = 0$, $x = \frac{\pi}{2}$ and the x -axis.

(i) Sketch R .

(ii) Find the exact volume of the solid generated when the region R is rotated about the x -axis.

(b) If α, β, γ , are the roots of the cubic polynomial equation $x^3 + 4x^2 - 6x - 8 = 0$ find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

3

(c) Find the term which is independent of x in the expansion of $\left(2x^3 + \frac{1}{3x^2} \right)^5$

3

(d) The remainder when $x^2 + ax + b$ is divided by $(x - 2)(x + 3)$ is $2x + 1$. Find the values of a and b .

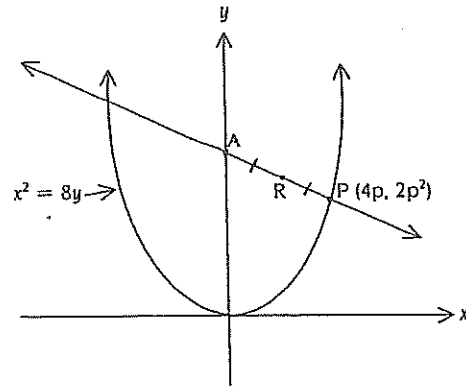
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Question Four (Start a NEW booklet)

Marks

7

(a)



$P(4p, 2p^2)$ is a variable point on the parabola $x^2 = 8y$ as shown in the diagram above.

The normal at P cuts the y -axis at A and R is the midpoint of AP .

- (i) Show that the normal at P has equation $x + py = 4p + 2p^3$
- (ii) Show that R has coordinates $(2p, 2p^2 + 2)$
- (iii) Show that the locus of R is a parabola and show that the vertex of this parabola is the focus of the parabola $x^2 = 8y$.

(b) (i) Evaluate $\int_1^3 \frac{dx}{x}$

(ii) Use Simpson's rule with 3 function values to approximate $\int_1^3 \frac{dx}{x}$

- (iii) Use your results to parts (i) and (ii) to obtain an approximation for e . Give your answer correct to 3 decimal places.

Question Five (Start a NEW booklet)

Marks

(a) Evaluate $\cos \left[\tan^{-1} \left(-\frac{1}{\sqrt{3}} \right) \right]$

1

- (b) When the temperature T of a certain body is 65°C it is cooling at the rate of 1°C per minute.

8

Assuming Newton's law of cooling: $\frac{dT}{dt} = -k(T - S)$ where

T is the temperature of the body at time t minutes

S is the temperature of the surrounding medium, assumed constant

k is a constant

- (i) Show that $T = S + Ae^{-kt}$ is a solution of the given differential equation, where A is also a constant.
- (ii) Show that the value of k is 0.02 given that S is 15°C .
- (iii) Find T when $t = 20$ minutes, giving your answer to the nearest degree. (You may assume that initially $T = 65$)
- (iv) How long will it take for the temperature of the body to fall to 35°C ?

- (b) The acceleration of a particle P , moving along a straight line has an acceleration given by

3

$$\frac{d^2x}{dt^2} = -4 \left(x + \frac{16}{x^3} \right)$$

Given that P is initially at rest at the point $x = 2$, show that the velocity v at any time is given by

$$v^2 = 4 \left(\frac{16 - x^4}{x^2} \right)$$

Question Six (Start a NEW booklet)

Marks

- (a) Prove by induction that, for all integers $n \geq 1$,

$$\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

3

- (b) Let $f(x) = x^2 + 6x$ for $x \geq -3$

6

- (i) Write down the range of $f(x)$.

- (ii) Briefly explain why the inverse function $f^{-1}(x)$ exists. Write down the domain and range of $f^{-1}(x)$.

- (iii) Find $f^{-1}(x)$. Sketch the graph of $y = f^{-1}(x)$.

3

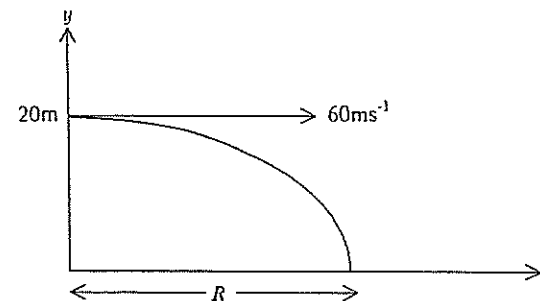
- (c) Sketch the graph of $y = 3 \cos^{-1}\left(\frac{x}{2} - 1\right)$.

Question Seven (Start a NEW booklet)

Ma

8

- (a)



An arrow is fired horizontally with a speed of 60ms^{-1} from the top of a 20m high wall on level ground as represented in the diagram above.

It is given that $\ddot{x} = 0$ and $\ddot{y} = -10$ where (x, y) is the position of the arrow at time t seconds after firing.

- Using calculus, show that $x = 60t$ and $y = 20 - 5t^2$.
- Find the time taken for the arrow to hit the ground.
- Find the distance R metres from base of the wall where the arrow hits the ground.
- Find the acute angle to the horizontal at which the arrow hits the ground.

- (b) It is given that:

$$(1+x)^{2n} = \sum_{k=0}^{2n} \binom{2n}{k} x^k$$

- (i) Show that $\sum_{k=0}^{2n} \binom{2n}{k} = 4^n$

$$(ii) \sum_{k=0}^{2n} \binom{2n}{k} \frac{1}{k+1} = \frac{4^{n+1} - 2}{4n+2}$$

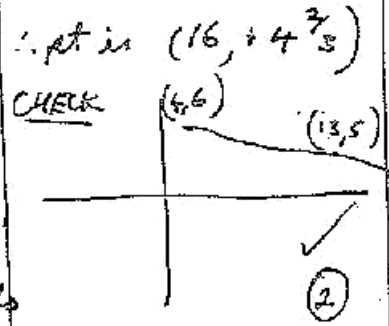
END OF EXAMINATION

(a) $\int_0^{2\sqrt{3}} \frac{dx}{4+x^2}$
 $= \left[\frac{1}{2} \tan^{-1} \frac{x}{2} \right]_0^{2\sqrt{3}}$ by table
 of standard integrals

$= \frac{1}{2} \tan^{-1} \sqrt{3} - 0$
 $= \frac{\pi}{6}$ (2)

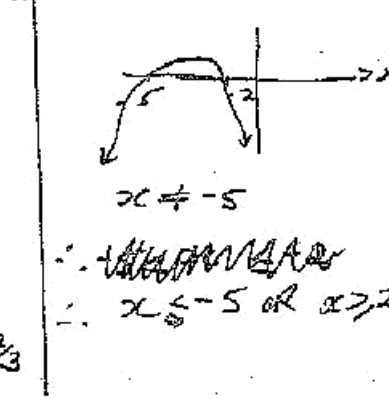
(b) $\frac{d}{dx} (\cos^3 x)$
 $= 3 \cos^2 x \cdot (-\sin x)$
 $= -3 \cos^2 x \sin x$ (2)

(c) $(4, 6)$ $\times$ 4
 $(13, 5)$ $\times$ -1
 Let (x_r, y_r) be this pt.
 $x_r = \frac{-4 + 52}{3}$
 $= \frac{48}{3} = 16$
 $y_r = \frac{-6 + 20}{3}$
 $= \frac{14}{3}$ or $4\frac{2}{3}$



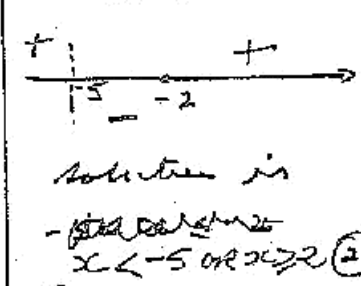
(d) $y = \frac{2x}{3x-1}$
 Vertical asymptote
 is $x = \frac{1}{3}$ (1)

(e) method 1
 $\frac{3}{x+5} \leq 1$ ($x \neq -5$)
 $3(x+5) \leq (x+5)^2$
 $(x+5)[3-(x+5)] \leq 0$
 $(x+5)(x-2) \leq 0$



Method 2 Sign diagram

$\frac{3}{x+5} \leq 1$
 $\Rightarrow \frac{3}{x+5} - 1 \leq 0$
 $\frac{-2-x}{x+5} \leq 0$
 $\frac{x+2}{x+5} \geq 0$



(f) $u = x^4$
 $du = 4x^3 dx$
 $\Rightarrow \frac{1}{2} du = 2x^3 dx$
 $x=0 \Rightarrow u=0$
 $x=\frac{1}{\sqrt{2}} \Rightarrow u=\frac{1}{4}$
 $I = \int_0^{\frac{1}{4}} \frac{1}{\sqrt{1-u}} \cdot \frac{1}{2} du$
 $= \left[-\sqrt{1-u} \right]_0^{\frac{1}{4}}$
 $= \sqrt{1} - \sqrt{\frac{3}{4}}$
 $= 1 - \frac{\sqrt{3}}{2}$ (2)

lim $\frac{\tan 2x}{4x}$
 $x \rightarrow 0$
 $= \frac{1}{2} \lim_{x \rightarrow 0} \frac{\tan 2x}{2x}$
 $= \frac{1}{2} \times 1$
 $= \frac{1}{2}$ (1)

(b) method 1
 Let $5\pi\theta + \sqrt{3} \cos\theta$
 $= R \sin(\theta + \alpha)$
 $R > 0$ $R \cos \alpha = \sqrt{3}$
 $R \sin \alpha = 5$
 $R^2 = 25 + 3 = 28$
 $R = 2\sqrt{7}$
 $\tan \alpha = \frac{5}{\sqrt{3}}$
 $\alpha = \arctan \frac{5}{\sqrt{3}}$

equation becomes
 $2 \sin \theta (\theta + \frac{\pi}{3}) = 1$
 $\sin(\theta + \frac{\pi}{3}) = \frac{1}{2}$
 Now $0 \leq \theta \leq 2\pi$
 $\Rightarrow \frac{\pi}{3} \leq \theta + \frac{\pi}{3} \leq 2\pi + \frac{\pi}{3}$

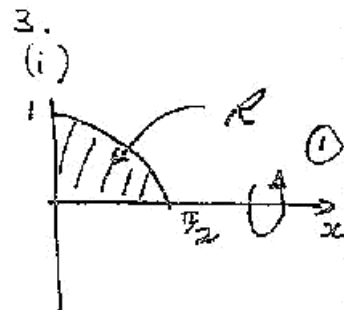
So $\theta + \frac{\pi}{3} = \frac{5\pi}{6}$ or $2\pi + \frac{\pi}{6}$
 $\therefore \theta = \frac{\pi}{2}$ or $\frac{11\pi}{6}$ (4)

method 2 t-formulas
 Let $x = \tan \theta/2$
 TEST $\theta = \pi$ (why?)
 LHS = 0 - 13
 RHS = 1
 $\theta \neq \pi$
 probably the more
 difficult method, but
 it can be done!

(c) $V = \frac{4}{3} \pi r^3$
 $\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$
 $= 4\pi r^2 \frac{dr}{dt}$
 $450 = 4\pi \times 15^2 \frac{dr}{dt}$
 $\frac{dr}{dt} = \frac{450}{4\pi \times 225}$
 $= \frac{1}{2\pi}$ or 0.159 cm s⁻¹ (3)

(d) (i) $f(0.5) = \cos 0.5$
 $= 1.57$
 $f(1.5) = \cos 1.5 = -0.334$
 $f(x)$ is continuous
 for $0.5 \leq x \leq 1.5$
 $\Rightarrow$ there exists a
 c in this open
 interval for $f(c)$
 (1)
 (ii) formula is
 $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$
 (or derive formula
 quickly)
 $x_1 = 1 - \frac{f(1)}{f'(1)}$
 $f(1) = \cos 1 = 0.54$
 $f'(1) = -\sin 1 = -0.84$
 $x_1 = 1 - \frac{0.54}{-0.84} = 1.64$

check is this a better approx
(Not required to show)
 $f(1) = \cos 1 > 0$
 $f'(x) = -\cos x + \frac{1}{x}$
 $f'(1) = -\cos 1 + 1 > 0$
So $x = 1.293$ is a better approx.



(ii)

$$V = \pi \int_0^{\frac{\pi}{2}} \cos^2 x \, dx$$

$$2 \cos^2 x = \cos 2x + 1$$

$$\therefore V = \pi \int_0^{\frac{\pi}{2}} (1 + \cos 2x) \, dx$$

$$= \pi \left[x + \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{2}}$$

$$= \pi \left(\frac{\pi}{2} + \frac{1}{2} \sin \pi \right)$$

$$= \frac{\pi^2}{2} \text{ unit}^3 \quad (2)$$

(b)

$$x^3 + 4x^2 - 6x - 8 = 0$$

$$\alpha + \beta + \gamma = -4$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = -6$$

$$\alpha\beta\gamma = 8$$

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$$

$$= \frac{-6}{8}$$

$$= -\frac{3}{4} \quad (3)$$

(c) $(2x^3 + \frac{1}{3x^2})^5$

method 1

general term

$$= \binom{5}{r} (2x^3)^{5-r} \left(\frac{1}{3x^2}\right)^r$$

$$= \binom{5}{r} \frac{2^{5-r}}{3^r} \frac{x^{15-3r}}{x^{2r}}$$

choose $r = 3$

$$\text{term} = \binom{5}{3} \frac{2^2}{3^3} x^{15-5}$$

$$= \frac{40}{27}$$

method 2

$$\left(2x^3 + \frac{1}{3x^2}\right)^5$$

$$= (x^3)^5 \left(2 + \frac{1}{3x^5}\right)^5$$

$$= x^{15} \left(2 + \frac{1}{3x^5}\right)^5$$

general term

$$= \binom{5}{3} \frac{2^2}{3^3} = \frac{40}{27}$$

(d) $x^3 + ax + b$

$$= (x-2)(x+3)Q(x) + 2x + 1$$

Let $P(x) = x^3 + ax + b$

$$P(2) = 8 + 2a + b = 2 \times 2 + 1$$

$$2a + b = -3$$

$$P(-3) = -27 - 3a + b = 2 \times -3 + 1$$

$$-3a + b = 22$$

Solving gives

$$a = -5$$

$$b = 7$$

4. (i)

$$x^2 = 8y \Rightarrow y = \frac{1}{8}x^2$$

$$y' = \frac{1}{4}x$$

M_{tangent} at $(4p, 2p^2)$

$$\text{is } \frac{1}{4} \times 4p = p$$

$$M_{\perp} = -\frac{1}{p}$$

$$y - 2p^2 = -\frac{1}{p}(x - 4p)$$

$$py - 2p^3 = -x + 4p$$

$$x + py = 4p + 2p^3 \quad (2)$$

(ii)

$$A = (0, 4 + 2p^2)$$

$(p \neq 0)$

$$R = \left(\frac{0+4p}{2}, \frac{4+2p^2+2p^2}{2}\right)$$

$$= (2p, 2p^2 + 2) \quad (1)$$

(iii)

$$\text{Let } x = 2p$$

$$y = 2p^2 + 2$$

$$= 2 \times \left(\frac{x}{2}\right)^2 + 2$$

$$= \frac{1}{2}x^2 + 2$$

$$\therefore y - 2 = \frac{1}{2}x^2$$

vertex

$\therefore$ vertex = $(0, 2)$

Path of $x^2 = 8y$

$r, x^2 = 4 \times 2 = y$

in focus = $(0, 2)$ (3)

if, Hence result.

(b) (i) $\int_1^3 \frac{dx}{x}$

$$= [\ln x]_1^3$$

$$= \ln 3 \quad (1)$$

(ii)

x	1	2	3
y	1	p_2	y_3

$$\int_1^3 \frac{dx}{x} \div \frac{1}{3} [y_1 + 4y_2 + y_3]$$

$$= \frac{1}{3} \left[1 + 4 \times \frac{1}{2} + \frac{1}{3} \right]$$

$$= \frac{10}{9} \quad (2)$$

(iii)

$$\ln 3 \div \frac{10}{9}$$

$$e^{\frac{10}{9}} = 3$$

$$(e^{\frac{10}{9}})^{\frac{9}{10}} = 3^{\frac{9}{10}}$$

$$e \div 3^{0.9}$$

$$= 2.68782 \dots$$

(c)

(i) LHS =

$$\frac{dT}{dt} = \frac{d}{dt} (5 + Ae^{-kt})$$

$$= 0 + A(-k)e^{-kt}$$

$$= -Ake^{-kt}$$

RHS =

$$-k(T - 5)$$

$$= -k(5 + Ae^{-kt} - 5)$$

$$= -Ake^{-kt}$$

(ii)

Newton's law

$$\frac{dT}{dt} = -k(T - 5)$$

$T = 65, \frac{dT}{dt} = 0$

$$0 = -k(65 - 5)$$

$$k = \frac{1}{50} = 0.02$$

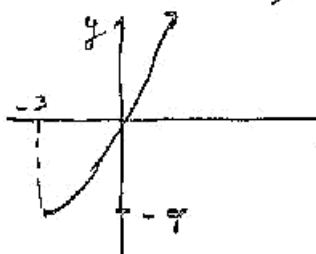
ii) $-0.02t$
 $T = 15 + Ae^{-0.02t}$
 When $t=0$ $T=65$
 $65 = 15 + A$
 $A = 50$

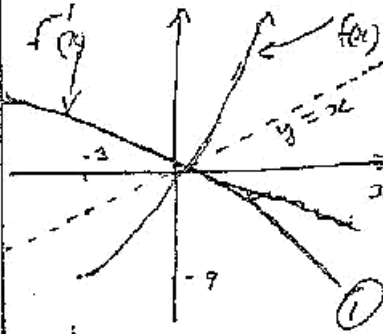
$T = 15 + 50e^{-0.02t}$
 $t = 20$
 $T = 15 + 50e^{-0.02 \times 20}$
 $= 15 + 50e^{-0.4}$
 $\approx 48.51 \dots$
 $\approx 49^\circ$ (2)

iv) $-0.02t$
 $T = 15 + 50e^{-0.02t}$
 $T = 35$ $t = ?$
 $35 = 15 + 50e^{-0.02t}$
 $e^{-0.02t} = \frac{2}{5}$
 $t = \frac{-1}{0.02} \ln 0.4$
 $= 45.81 \dots$
 ≈ 46 minutes (3)

(b) $\frac{d^2x}{dt^2} = -4(x + \frac{16}{x^3})$
 $\frac{d^2x}{dt^2} = \frac{d}{dx}(\frac{1}{2}v^2)$
 $\frac{d}{dx}(\frac{1}{2}v^2) = -4(x + \frac{16}{x^3})$
 $\frac{1}{2}v^2 = -4(\frac{1}{2}x^2 + \frac{16}{-2}x^{-2}) + C$
 $v^2 = -4(x^2 - \frac{16}{x^2}) + C'$
 $v = 0, x = 2$
 $0 = -4(2^2 - \frac{16}{2^2}) + C'$
 $= 0 + C'$
 $\Rightarrow C' = 0$
 $\therefore v^2 = -4(x^2 - \frac{16}{x^2})$
 $= 4(\frac{16 - x^4}{x^2})$ (2)

6. (a) $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$
 $f(n) = 1$
 $LHS = \frac{1}{1 \times 2} = \frac{1}{2}$
 $RHS = \frac{1}{1+1} = \frac{1}{2}$
 $\therefore$ true for $n=1$
Suppose Statement true for $n=k$
 $\frac{1}{1 \times 2} + \dots + \frac{1}{k(k+1)} = \frac{k}{k+1}$
 Consider $n = k+1$
 $\frac{1}{1 \times 2} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)}$
 $= \frac{k}{k+1} + \frac{1}{(k+1)(k+2)}$
 $= \frac{k(k+2) + 1}{(k+1)(k+2)}$
 $= \frac{(k+1)^2}{(k+1)(k+2)}$
 $= \frac{k+1}{k+2}$
 So result true for $n=k$ Proved
 true for $n=k+1$
 $\therefore$ by M.I. true for $n=1, 2, 3, \dots$

5(a) (i) $f(x) = x(x+6)$

 Range: $f(x) \geq -9$ (1)
 (ii) $f(x)$ exists as for $x \geq -3$ $f(x)$ is increasing. (1)
 Domain of $f(x)$ is $x \geq -9$ (1)
 (iii) $y = x(x+6)$
 $= x^2 + 6x$
 $x \geq -3, y \geq -9$
 $f(x): x \leftrightarrow y$
 $x = y^2 + 6y$
 $x \geq -9$ and $y \geq -3$
 $y^2 + 6y - x = 0$

$y = \frac{-6 \pm \sqrt{36 - 4x}}{2}$
 $= -3 \pm \sqrt{9 - x}$
 but $y \geq -3$ f
 $\therefore y = -3 + \sqrt{9 - x}$
 $\Rightarrow f(x) = -3 + \sqrt{9 - x}$ (2)

 (c) $y = 3 \cos^{-1}(\frac{x}{2} - 1)$
Domain
 $-1 \leq \frac{x}{2} - 1 \leq 1$
 $0 \leq \frac{x}{2} \leq 2$
 $0 \leq x \leq 4$
Range
 $0 \leq y \leq 3\pi$


7 (i) $\ddot{x} = 0$
 $\therefore \dot{x} = C$
 at $t=0$ $\dot{x} = 60$
 $\therefore \dot{x} = 60$
 $x = 60t + C$
 at $t=0, x=0$
 $\Rightarrow C=0$
 $\therefore x = 60t$
 $\ddot{y} = -10$
 $\dot{y} = -10t + C$
 at $t=0, \dot{y} = 0$
 $\Rightarrow C=0$
 $\therefore \dot{y} = -10t$
 $y = -5t^2 + C$
 at $t=0, y = 20$
 $\Rightarrow C=20$
 $\therefore y = 20 - 5t^2$
 (ii) at ground $(x, y) = (R, 0)$
 Solve $y = 0$
 $20 - 5t^2 = 0$
 $t^2 = 4$
 $t > 0 \Rightarrow t = 2$
 (iii) at $(R, 0)$
 $\therefore x(2) = 60 \times 2 = 120$

v) at ground
vector origin



$k=2$

$$\dot{y} = -10 \times 2$$

$$= -20$$

$$\dot{x} = 60$$

$$\tan \theta = \left| \frac{\dot{y}}{\dot{x}} \right|$$

$$= \frac{20}{60}$$

$$= \frac{1}{3}$$

$$\theta = \tan^{-1} \frac{1}{3} \quad (2)$$

$$\approx 18.43^\circ$$

$$\underline{\underline{18^\circ}}$$

(b)

$$(1+x)^{2n} = \sum_0^{2n} \binom{2n}{k} x^k$$

(i) put $x=1$

$$2^{2n} = \sum_0^{2n} \binom{2n}{k} 1^k$$

$$4^n = \sum_0^{2n} \binom{2n}{k}$$

(1)

(ii) integrating

$$\frac{(1+x)^{2n+1}}{2n+1} + C = \sum_0^{2n} \binom{2n}{k} \frac{x^{k+1}}{k+1}$$

put $x=0$

$$\frac{1}{2n+1} + C = 0$$

$$\Rightarrow C = -\frac{1}{2n+1}$$

$$\therefore \sum_0^{2n} \binom{2n}{k} \frac{x^{k+1}}{k+1}$$

$$= \frac{(1+x)^{2n+1}}{2n+1} - \frac{1}{2n+1}$$

put $x=1$

$$\sum_0^{2n} \binom{2n}{k} \frac{1}{k+1}$$

$$= \frac{2^{2n+1} - 1}{2n+1}$$

$$= \frac{2 \times 4^n - 1}{2n+1} \times \frac{2}{2}$$

$$= \frac{4 \times 4^n - 2}{4n+2}$$

$$= \frac{4^{n+1} - 2}{4n+2}$$

(3)